Project 01 - Simplex

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1 History of the Simplex Method

- George B. Dantzig developed the Simplex method in 1947 to solve linear programming problems.
- The method seeks an optimal solution among the basic feasible solutions, associated with the vertices of the feasible region.
- In each iteration, a variable enters the base and another leaves, resulting in a new basic feasible solution. This is known as pivot operation.
- In bounded problems, iterations advance toward an improvement of the objective; in degenerate cases, the value may remain the same.
- It can detect optimality, unboundedness, and, according to the formulation, infeasibility.
- It is efficient in many practical problems, although its worst-case theoretical complexity is exponential.

2 Original Problem

Problem Name: Geppetto's Toy Factory

$$
\begin{array}{rcl}
\max \quad Z & = & 3\,\texttt{toy_soldiers} + 2\,\texttt{toy_trains} \\
\text{subject to} & & \\
2\,\texttt{toy_soldiers} + 1\,\texttt{toy_trains} & \leq & 100 \\
1\,\texttt{toy_soldiers} + 1\,\texttt{toy_trains} & \leq & 80 \\
1\,\texttt{toy_soldiers} & \leq & 40 \\
\texttt{toy_soldiers} & \geq & 0 \\
\texttt{toy_trains} & \geq & 0
\end{array}
$$

3 Tabla Simplex inicial

Row	Z	toy_soldiers	toy_trains	S_1	S_2	S_3	RHS
Z	1	-3	-2	0	0	0	0
R_1	0	2	1	1	0	0	100
R_2	0	1	1	0	1	0	80
R_3	0	1	0	0	0	1	40

4 Tabla despues del paso 1

Row	Z	toy_soldiers	toy_trains	S_1	S_2	S_3	RHS
Z	1	0	-2	0	0	3	120
R_1	0	0	1	1	0	-2	20
R_2	0	0	1	0	1	-1	40
R_3	0	1	0	0	0	1	40

5 Tabla despues del paso 2

Row	Z	toy_soldiers	toy_trains	S_1	S_2	S_3	RHS
Z	1	0	0	2	0	-1	160
R_1	0	0	1	1	0	-2	20
R_2	0	0	0	-1	1	1	20
R_3	0	1	0	0	0	1	40

6 Tabla despues del paso 3

Row	Z	toy_soldiers	toy_trains	S_1	S_2	S_3	RHS
Z	1	0	0	1	1	0	180
R_1	0	0	1	-1	2	0	60
R_2	0	0	0	-1	1	1	20
R_3	0	1	0	1	-1	0	20

7 Problem Solution

- An optimal solution was found.
- **Optimal Solution**
  - Objective Function Value: $Z = 180$.
  - `toy_soldiers` $= 20$ (basic)
  - `toy_trains` $= 60$ (basic)
  - $S_1 = 0$ (non-basic)
  - $S_2 = 0$ (non-basic)
  - $S_3 = 20$ (basic)

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  - $S_2 = 0$ (non-basic)
  - $S_3 = 20$ (basic)